- Custom Namespace structure

College Library

- CollegeLibrary . Models
  - Books
  - Students
  - Librarian
  - Issued Transaction
- Collegelibrary . Data
  - Database Connection
  - Book Repository
  - Student Repository
- College Library - Services
  - BookService
  - Student Service
  - IssueService
  - Fine Service
- College Library . UI
  - Login Form
  - Book Form
  - Student Form
- College library - Utilities
  - Validation
  - Logger

1) CLR (Common Language Runtime):
It performs several important tasks:
- Memory management
- Garbage collection
- Exception handaling
- Security
- Thread management
- Type checking

2) FCL (Framework class Library):
Provides reusable classes that can be used to develop the library system:
- Collection of books → List<T>
- File operations → System.IO
- Database connectivity → ADO.NET
- Date & time → DateTime
- Exception handling → System.Exception
- Text processing → System.String
- Security-related functionality → .NET security APIs

3) JIT (Just in time):

C# Source Code → C# Compiler → Intermediate Language
↓
Native Machine code ← JIT Compiler ← CLR
↓
CPU

- Private assemblies should be used for application-specific modules, while a shared / public assembly can be used when a common library needs to be shared accross multiple applications

Fully Qualified Names:

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- A fully qualified class name include its namespace.
- Unqualified class names can only be used in scope.

→ Suppose two different namespaces contains classes with the same name, we can resolve the conflict by using their fully qualified names:

College - Student collegeStudent = new College. Student();
School . Student schoolStudent = new School. Student ();

Now there is no ambiguity because the complete namespace is specified.

Question 6: Case Study: A college libraries wants to modernize its software using the .NET Framework. The application should be easy to maintain, reusable & support future expansion.

- Explaining why the .NET Framework is suitable.
- Identify CLR, FCL, & JIT components used.
- Choosing between private & public assemblies.
- Create a custom namespace structure.

→ Case Study: Modernizing a college Library using .NET Framework.

- The .NET Framework is suitable for developing such as application because it provides a managed runtime environment, reusable class libraries, support for multiple programming languages, databases connectivity, security features, & a well-organized architecture.

Question 5) Explain namespaces in C#. Discuss predefined, custom, & nested namespaces with examples. What is a fully qualified name? Explain how it resolves name conflicts. (4)

→ Namespaces are a way to define the classes & other types of information into one hierarchical structure.

1) Predefined namespace: provided by the .NET Framework & the C# libraries. They contains commonly used classes & types. Example:

1) System: Fundamental types & functionality (Console, string, Math)

2) System.Collection.Generic: Generic collection (List<T>, Dictionary<TKey, TValues>)

2) Custom namespace: also called used defined namespace is created by the programmer to organize application-specific types.

Example:
```
namespace College {
    class Student {
        public void Display(){
            Console.WriteLine('Abc');
        }
    }
}
```

3) Nested Namespaces: declared inside another namespace. It is useful for creating a hierarchical organization of related types.

Example:
```
namespace College {
    namespace Department {
        class Student {
            public void Display(){
                Console.WriteLine('Abc');
            }
        }
    }
}
```

Question 3) Explain the Just-in-time (JIT) compiler & discuss its different types

→ The Just-in-time (JIT) compiler is an important component of the Common language Runtime (CLR) in the .NET framework. Its main purpose is to convert Intermediate language (IL) code into native machine code that can be executed directly by the computer processor.

Types of JIT Compiler:

1) Pre-JIT Compiler: compiles the entire assembly into native machine code before the application begins execution.

2) Normal-JIT Compiler: compiles methodes when they are called for the first time.

3) Econo-JIT Compiler: was designed for environments are where memory usage are particular important. It compiled methods as needed & could discarded generated native code when memory was required.

Question 4) Differentiate between private assemblies & public assemblies.

→ In the .NET framework, a private assembly is an assembly that is intended to be used by a single application. These assemblies are typically stored in the application's directory or in a subdirectory, making them specific to that application

→ In the .NET framework, a shared or public assembly is ~~..~~ intended to be used by multiple applications. Unlike private assembly, public assembly are typically stored in the Global Assembly Cache (GAC), making them accessible globally on a system.

Main Components:

Common Language Runtime (CLR): Basically, it is responsible for managing the execution of .NET programs regardless of any .NET programming language.

Framework Class Library (FCL): It is the collection of reusable, object-oriented class libraries & method, etc that can be integrated with CLR. Also called the Assembles.

Question 2) Describe the role of the Common Language Runtime (CLR) & Framework Class Library (FCL). Differentiate b/w managed & unmanaged code.

→ Common Language Runtime (CLR): is the basic & Virtual Machine Component of the .NET Framework. It is the run-time environment in the .NET Framework that runs the codes & helps in making the development process easier by providing various services such as remoting, thread management, type safety, memory management, robustness, etc.

Framework Class Library (FCL): It is the collection of reusable, object oriented class libraries & methods, etc that can be integrated with CLR.

Managed Code runs under the control of the CLR (Common Language Runtime), while unmanaged code runs directly on the operating system

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Assignment-1

Question 1). Explain the evolution of the .NET Framework & its major versions. Draw & explain the architecture of the .NET Framework.

→ The .NET Framework evolved through several major versions, with each versions adding new features, improving performance & expanding application development capabilities

Major Versions:

- .NET Framework (1.0) : 2002 : First official release, introduced CLR, CTS, CLS, ASP.NET & Windows Forms
- .NET Framework (4.8.1) : 2022 : Latest major .NET framework release; improved support for modern Windows & accesibility